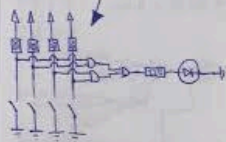


Logic gates Problems 8

A	B	C	D	S ₁	S ₂
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	0	1
0	0	1	1	0	1
0	1	0	0	0	1
0	1	0	1	0	1
0	1	1	0	0	1
0	1	1	1	0	1
1	0	0	0	1	0
1	0	0	1	1	0
1	0	1	0	1	0
1	0	1	1	1	0
1	1	0	0	1	0
1	1	0	1	1	0
1	1	1	0	1	0
1	1	1	1	1	0

$$S_1 = A \cdot B \cdot C \cdot D$$



$$S_2 = (\bar{A} \cdot \bar{B} \cdot \bar{C} \cdot \bar{D}) + (\bar{A} \cdot \bar{B} \cdot \bar{C} \cdot D) + (\bar{A} \cdot \bar{B} \cdot C \cdot \bar{D}) + (\bar{A} \cdot \bar{B} \cdot C \cdot D) + (\bar{A} \cdot B \cdot \bar{C} \cdot \bar{D}) + (\bar{A} \cdot B \cdot \bar{C} \cdot D) + (\bar{A} \cdot B \cdot C \cdot \bar{D}) + (\bar{A} \cdot B \cdot C \cdot D) + (A \cdot \bar{B} \cdot \bar{C} \cdot \bar{D}) + (A \cdot \bar{B} \cdot \bar{C} \cdot D) + (A \cdot \bar{B} \cdot C \cdot \bar{D}) + (A \cdot \bar{B} \cdot C \cdot D) + (A \cdot B \cdot \bar{C} \cdot \bar{D}) + (A \cdot B \cdot \bar{C} \cdot D) + (A \cdot B \cdot C \cdot \bar{D}) + (A \cdot B \cdot C \cdot D)$$

A: escáner biométrico de investigador autorizado = 1
 B: cabina bien despresurizada = 1
 C: parada en octóvada = 1
 D: lector de tarjetas = 1

$$S_2 = \bar{B} + \bar{C}$$

AB	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	1	0	0	1
10	1	0	0	1

